Data Science & AI using Machine Learning, Deep Learning & NLP

90 Hours Industry-Oriented Training Program

Training Duration: 90 Hours

Module 1: Python Programming for Data Science

Topics Covered

- Introduction to Python
- Variables and Data Types
- Operators and Expressions
- Conditional Statements
- Loops and Iterations
- Functions and Modules
- File Handling
- Exception Handling
- Object-Oriented Programming Basics
- NumPy Fundamentals
- Pandas Fundamentals

Practical Assignments

- Student Management System
- Sales Data Analysis
- Data Cleaning Exercises

Learning Outcome

- Write Python programs independently
- Use NumPy and Pandas for data manipulation
- Understand coding standards used in Data Science

Module 2: Data Analysis & Visualization

Topics Covered

- Data Collection Techniques
- Data Preprocessing
- Missing Value Treatment
- Outlier Detection
- Exploratory Data Analysis (EDA)
- Data Transformation
- Feature Engineering Basics

Visualization Tools

- Matplotlib
- Seaborn
- Plotly

Charts Covered

- Line Chart
- Bar Chart
- Histogram
- Scatter Plot
- Box Plot
- Heatmap
- Correlation Matrix

Mini Project

- E-Commerce Sales Dashboard

Learning Outcome

- Analyze raw datasets
- Create professional visualizations
- Extract business insights

Module 3: Statistics & Mathematics for Machine Learning

Topics Covered

- Descriptive Statistics
- Probability Theory
- Normal Distribution
- Hypothesis Testing
- Correlation
- Covariance
- Linear Algebra Basics
- Vectors & Matrices
- Gradient Descent Concept

Practical Exercises

- Statistical Analysis using Python
- Business Dataset Interpretation

Learning Outcome

- Understand mathematical foundations of ML
- Apply statistical concepts in real datasets

Module 4: Machine Learning Fundamentals

Introduction to Machine Learning

Supervised Learning

- Regression vs Classification
- Training & Testing Dataset
- Model Evaluation

Algorithms Covered

Regression Models

- Linear Regression

- Multiple Linear Regression
- Polynomial Regression

Classification Models

- Logistic Regression
- Decision Tree
- Random Forest
- KNN
- Support Vector Machine (SVM)
- Naive Bayes

Clustering

- K-Means Clustering
- Hierarchical Clustering

Model Evaluation

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Mini Projects

1. House Price Prediction
2. Employee Salary Prediction
3. Customer Segmentation
4. Loan Approval Prediction

Learning Outcome

- Build predictive models
- Evaluate model performance
- Solve real-world business problems

Module 5: Deep Learning using TensorFlow & Keras

Introduction to Deep Learning

Artificial Neural Networks (ANN)

- Biological vs Artificial Neurons
- Perceptron
- Activation Functions
- Feed Forward Networks

TensorFlow & Keras

- TensorFlow Basics
- Building Neural Networks
- Model Compilation
- Training & Validation

Deep Learning Concepts

- Backpropagation
- Optimizers
- Loss Functions
- Hyperparameter Tuning

Convolutional Neural Networks (CNN)

- Image Processing
- Feature Extraction
- Image Classification

Recurrent Neural Networks (RNN)

- Sequence Prediction
- Time Series Forecasting

LSTM Networks

- Sequential Data Processing

Deep Learning Projects

1. Handwritten Digit Recognition
2. Face Mask Detection
3. Image Classification System
4. Stock Price Prediction

Learning Outcome

✓ Build neural networks

✓ Train image classification models

✓ Understand advanced AI concepts

Module 6: Natural Language Processing (NLP)

Introduction to NLP

Text Processing

- Tokenization
- Stop Word Removal
- Lemmatization
- Stemming

NLP Libraries

- NLTK
- SpaCy
- TextBlob

Feature Extraction

- Bag of Words
- TF-IDF
- Word Embeddings

NLP Applications

- Sentiment Analysis
- Text Classification
- Spam Detection
- Chatbot Development

Introduction to Transformers

- BERT
- GPT Models
- Hugging Face Basics

NLP Projects

1. Movie Review Sentiment Analysis
2. Spam Email Detection
3. Resume Screening System
4. AI Chatbot

Learning Outcome

- Process and analyze text data
- Build NLP applications
- Understand Generative AI foundations

Module 7: Industry Capstone Project & Deployment

Project Development Lifecycle

Steps

- Problem Identification
- Dataset Collection
- Data Cleaning
- Model Building
- Model Evaluation
- Deployment

Deployment Tools

- Streamlit
- Flask
- GitHub

Dashboard Integration

- Interactive AI Applications
- Visualization Reports

Major Capstone Project Options

Project 1: AI-Powered Student Performance Predictor

Technologies: Python, ML, Streamlit

Features

- Predict academic performance
- Attendance analysis
- Risk prediction

Project 2: E-Commerce Recommendation System

Technologies: Python, ML, NLP

Features

- Product recommendations
- User behavior analysis
- Personalized suggestions

Project 3: Fake News Detection System

Technologies: NLP, Machine Learning

Features

- News authenticity prediction
- Text classification
- Real-time analysis

Project 4: AI Resume Screening System

Technologies: NLP, Deep Learning

Features

- Resume ranking
- Skill matching
- Candidate recommendation

Project 5: Customer Churn Prediction

Technologies: Machine Learning

Features

- Customer retention prediction
- Business insights dashboard

Project 6: AI-Based Medical Diagnosis Assistant

Technologies: Machine Learning, Deep Learning

Features

- Disease prediction
- Symptom analysis
- Healthcare insights

Tools & Technologies Covered

- Python
- Jupyter Notebook
- Google Colab
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-Learn
- TensorFlow
- Keras
- NLTK
- SpaCy
- Hugging Face
- Streamlit
- Flask
- GitHub
- VS Code

Deliverables to Students

- Training Notes & PPTs
- Source Codes
- Datasets
- Assignments
- Mini Projects
- Major Project
- Completion Certificate

Program Outcomes

After completing this 90-hour training, students will be able to:

Technical Skills

- Perform Data Analysis and Visualization
- Develop Machine Learning Models
- Build Deep Learning Applications

- Create NLP-based Solutions
- Deploy AI Applications
- Work with Real-World Datasets

Career Readiness

- Industry-Level Project Experience
- Portfolio Development
- Problem-Solving and Analytical Skills
- Foundation for Data Scientist, ML Engineer, AI Engineer, NLP Engineer, and Data Analyst roles